
High Fever, Low Blood Pressure

Dr CHAN Wai Ming
Adult Intensive Care Unit
Queen Mary Hospital



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High Fever, Low Blood Pressure

- ▶ Definition of Sepsis
- ▶ Multi-Organ Failure
- ▶ Treatment of Sepsis
 - ▶ Surviving Sepsis Campaign Guidelines
 - ▶ Targets of Resuscitation
 - ▶ Use of Vasopressors
 - ▶ Use of Fluids

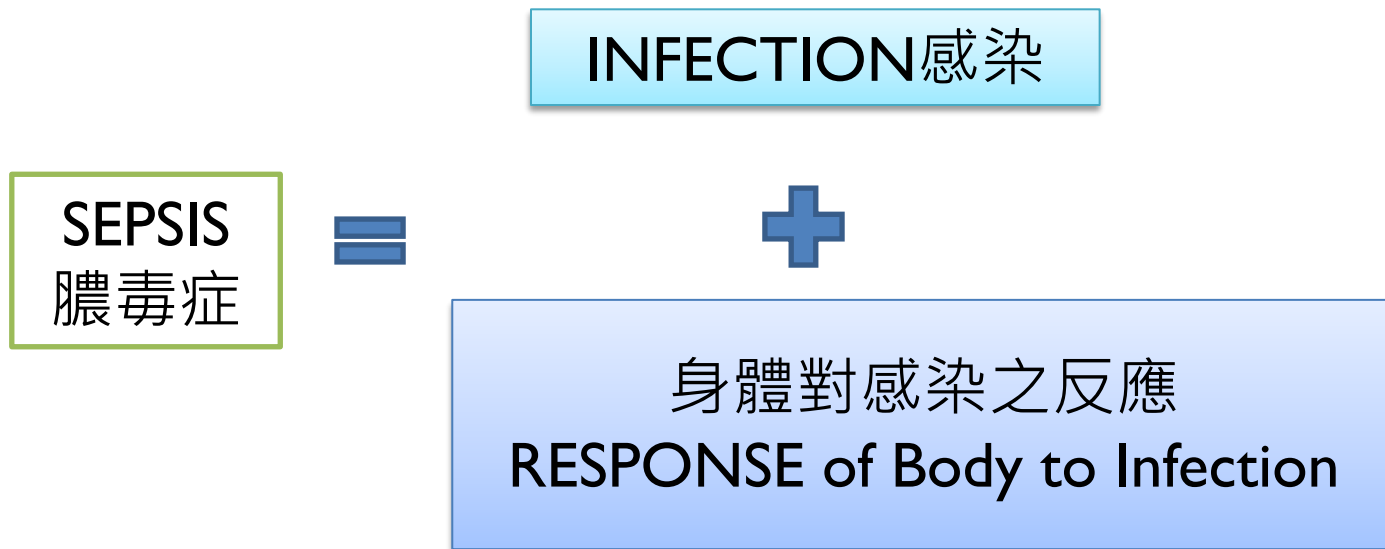
How Common is Sepsis?

- ▶ European Multinational Study (SOAP) Crit Care Med 2006
- ▶ Year 2002, 195 ICU in 24 Countries
 - ▶ 2 wk cross-sectional study involving 3,147 pts
 - ▶ 37.4% ICU patients had sepsis
 - ▶ Shock 1058 (33.6%)
 - ▶ 462(14.7%) Septic shock
 - ▶ Hospital Mortality: 24%
- All Septic patients in ICU in China (*J Xie et al, CCM 2019*)
- 44 Hospitals, 01/12/15-31/01/16), n=2322
- 33.9% hospital acquired sepsis
- =20.6/100 ICU admissions
- D90 Mortality 35.5%

The Old Concept of SIRS:

~~Systemic Inflammatory Response Syndrome~~

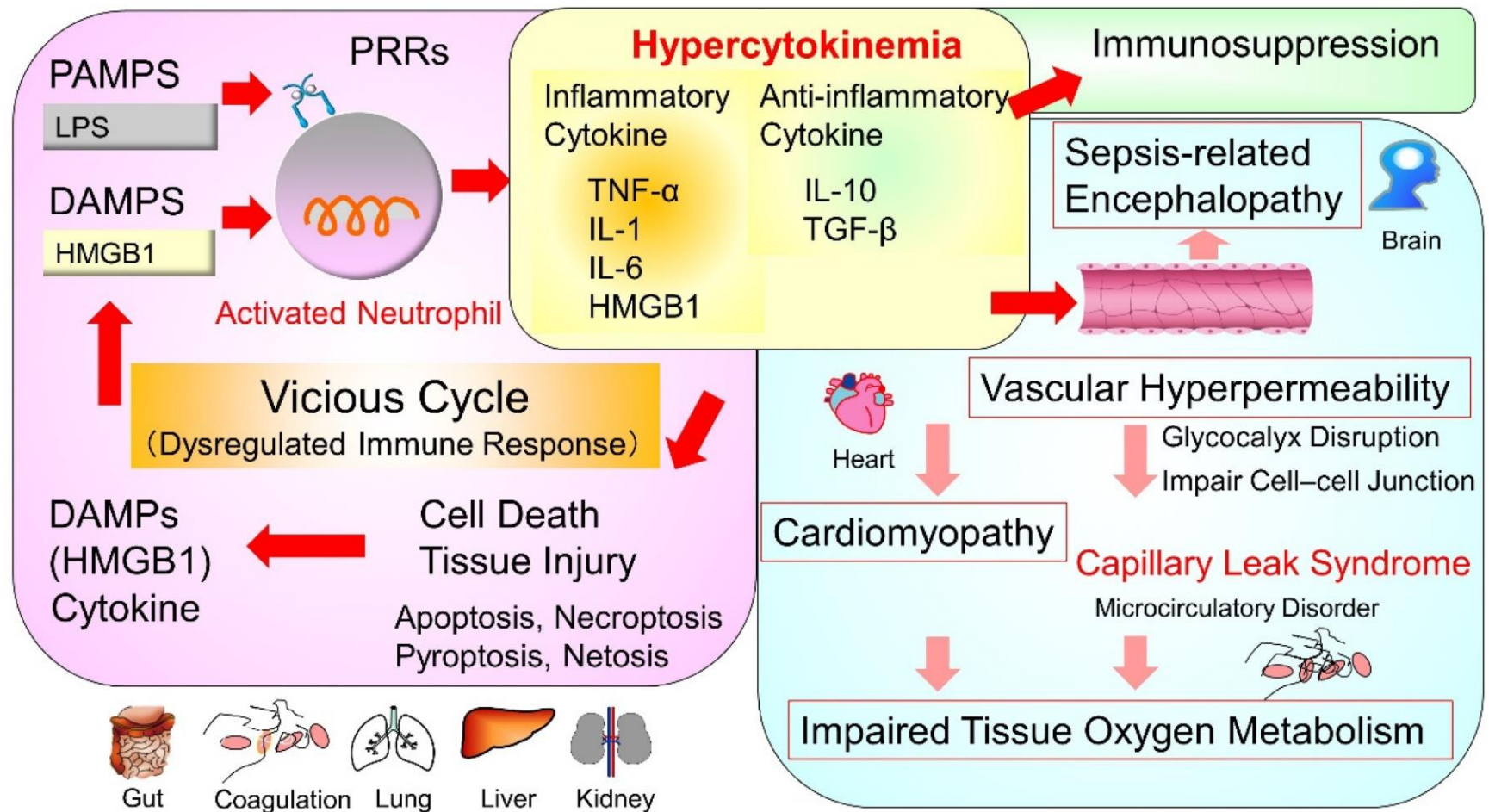
- ▶ Based on Definition in 1992
 - ▶ The **Systemic** activation of **Innate Immune Response** to Injury **regardless** of the cause.
- ▶ Definition after 2001: SIRS Not Obligatory



What is INFECTION?

- ▶ A **microbial** phenomenon
- ▶ Defined as the invasion of pathogens
 - ▶ Survival of the pathogen in sterile space cases:
 - ▶ central nervous system, lung, pericardium pathogens
 - ▶ survival to anomaly space cases:
 - ▶ tuberculosis, influenza virus obvious
 - ▶ clinical or pathological infection characterization
 - ▶ e.g: abscess, endocarditis
- ▶ No Commonly accepted Serum Marker of sepsis
 - ▶ **Procalcitonin** promising but not gold standard.

A Complicated Response



SEPSIS-3 Guideline 2016

- Definition: life-threatening organ dysfunction caused by a dysregulated host response to infection.
 - Septic Shock
 - Subset of sepsis with circulatory and cellular/metabolic dysfunction associated with higher risk of mortality
 - Clinically identified by
 - a vasopressor requirement to maintain a mean arterial pressure of 65 mm Hg or greater and serum lactate level greater than 2 mmol/L (>18 mg/dL) in the absence of hypovolemia.
-



Multi-Organs Dysfunction Syndrome (MODS) Due to Sepsis

Manifestation of Septic Shock is normally **Stereotypic**

- ▶ **Circulatory System**
 - ▶ Distributive Shock

Types of Shock

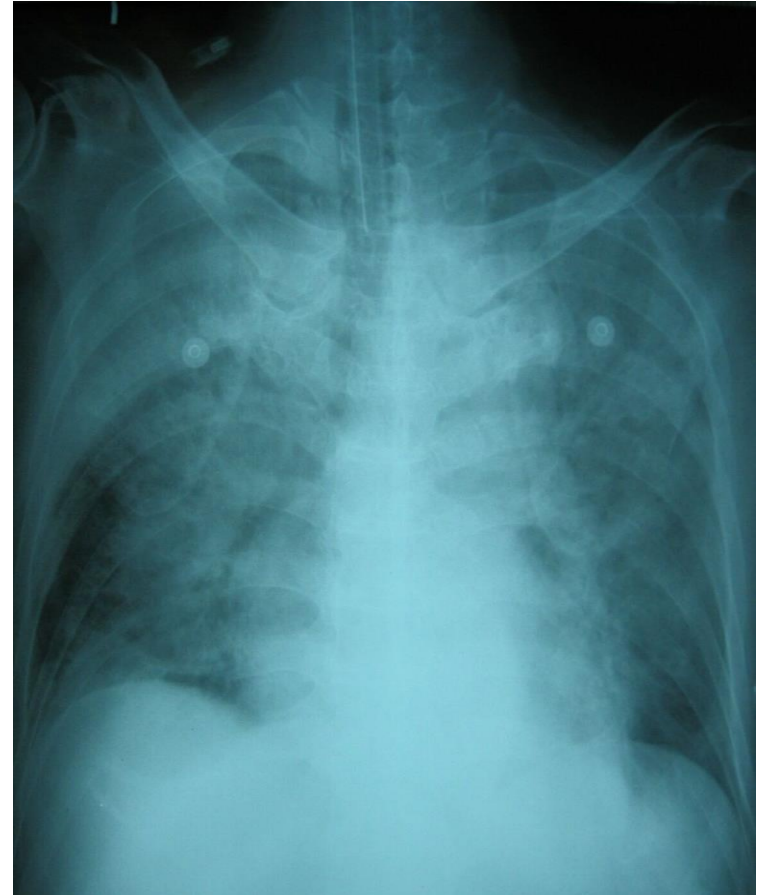
	Distributive	Cardiogenic	Hypo volaemic	Obstructive
Blood Pressure	↓	↓	↓	↓
Cardiac Output	↑	↓	↓	↓
Preload	↑ → ↓	↑	↓	↑
Systemic Vascular Resistance (Afterload)	↓	↑	↑	↑

Typical Lung Injury

- ▶ **Acute Respiratory Distress Syndrome (ARDS)**
- ▶ Non-Cardiogenic Pulmonary Oedema
 - ▶ Type I Failure
 - ▶ Low Lung Compliance

Euro-American Consensus Definition: ARDS 1993

- ▶ Acute onset with known insults
- ▶ All 3 Criteria need to be met:
 - ▶ CXR: Bilateral Airspace Shadows compatible with Pulmonary oedema
 - ▶ No evidence of L Atrial Hypertension or PAWP < 18mmHg
 - ▶ $\text{PaO}_2/\text{FiO}_2 < 200\text{mmHg}$ ($\approx 26.7\text{KPa}$)
 - ▶ Acute Lung Injury (ALI) if $\text{PaO}_2/\text{FiO}_2 < 300\text{mmHg}$ ($\approx 40\text{KPa}$)



PAWP: Pulmonary Arterial Wedge Pressure

ARDS: Berlin Definition 2012

“Backward Compatibility” such that data from the AECC definition could still be useful

Timing	Within 1 Week of Known Clinical Insult or New Resp Symptom
Chest Imaging	Bilateral Opacities: Not fully explained by effusions, lobar/lung collapse or nodules
Origin of edema	Not fully explained by cardiac failure or fluid overload Need objective assessment if no Risk Factor
Oxygenation	
Mild	$200 \text{ mm Hg} < \text{PaO}_2/\text{FIO}_2 \leq 300 \text{ mm Hg}$ with PEEP or CPAP $\geq 5 \text{ cm H}_2\text{O}$
Moderate	$100 \text{ mm Hg} < \text{PaO}_2/\text{FIO}_2 \leq 200 \text{ mm Hg}$ with PEEP $\geq 5 \text{ cm H}_2\text{O}$
Severe	$\text{PaO}_2/\text{FIO}_2 \leq 100 \text{ mm Hg}$ with PEEP $\geq 5 \text{ cm H}_2\text{O}$

JAMA. 2012;307(23):2526-2533. doi:10.1001/jama.2012.5669.

RIFLE Criteria

(ADQI workgroup Crit Care 2004)

Proposed classification scheme for acute renal failure

	GFR criteria	Urine output criteria	
Risk	Increased SCreat x1.5 or GFR decrease >25 percent	UO <.5 mL/kg/h x 6 hr	
Injury	Increased SCreat x2 or GFR decrease >50 percent	UO <.5 mL/kg/h x 12 hr	
Failure	Increase SCreat x3 GFR decrease 75 percent OR SCreat ≥4 mg/dL Acute rise ≥0.5 mg/dL	UO <.3 mL/kg/h x 24 hr or Anuria x 12 hrs	High sensitivity <i>Oliguria</i>
Loss	Persistent ARF = complete loss of kidney function >4 weeks		
ESKD	End stage kidney disease (>3 months)		High specificity

MODS related to Sepsis

Organ System	Pathology	Clinical Manifestation
CVS	Septic Shock	High Cardiac Output, Vasodilatation
Resp	ARDS	Type I Failure
Renal	ATN	Oliguric Renal Failure
Blood	DIC	↓ Platelets
GI	Multi-factorial	GI Bleeding, Ileus
Liver	Multi-factorial	Jaundice
CNS	Septic Encephalopathy	Encephalopathy
Peri NS	Critical Illness Polyneuropathy	Weakness, Weaning Failure

Treatment of Sepsis in 2021

- ▶ **Surviving Sepsis Campaign 2021 Guideline**
 - ▶ Infection Source Control
 - ▶ Eradication of Source if possible
 - ▶ Prompt Initiation of Appropriate Antibiotics
 - ▶ Appropriate Fluid Resuscitation
 - ▶ Favour Crystalloids over colloids
 - ▶ Weak Recommendation of using Balanced Solution
 - ▶ Appropriate use of Vasopressors
 - ▶ Preferred Vasopressor: **Noradrenaline**
 - ▶ Other Adjunctive Therapy
 - ▶ Appropriate Ventilatory Strategy
 - ▶ Appropriate use of Renal Replacement Therapy
 - ▶ Appropriate Nutritional Protocol

<http://www.survivingsepsis.org>

Importance of Prompt and Correct Antibiotics Treatment

Kumar et al *Crit Care Med* 2006

- ▶ Retrospective Review 2731 Pts with Septic Shock in 14 US/Canada ICUs July 1989 and June 2004

Each **hour of delay** in antimicrobial Rx over the ensuing 6 hrs was associated with an average **decrease** in survival of **7.6% / hr.**

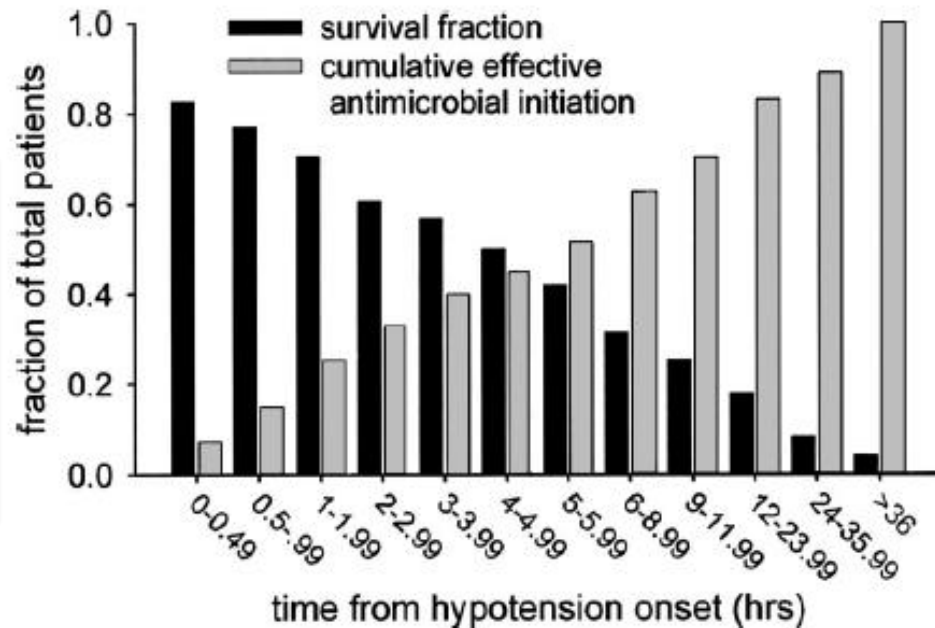


Figure 1. Cumulative effective antimicrobial initiation following onset of septic shock-associated hypotension and associated survival. The x-axis represents time (hrs) following first documentation of septic shock-associated hypotension. *Black bars* represent the fraction of patients surviving to hospital discharge for effective therapy initiated within the given time interval. The *gray bars* represent the cumulative fraction of patients having received effective antimicrobials at any given time point.

Presumed Hemodynamics Targets of Perfusions to Achieve

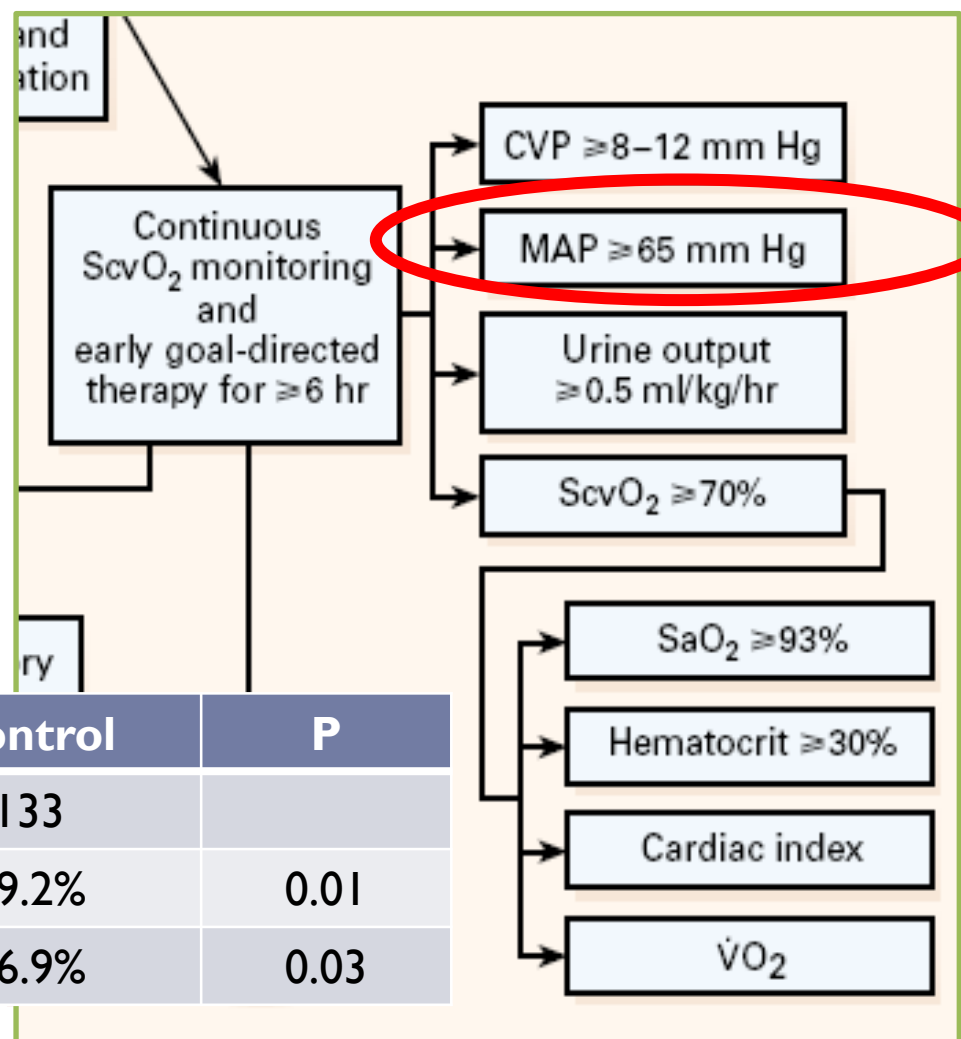
- ▶ Targets, as derived from **Early Goal Directed Therapy**
 - ▶ Central Venous Pressure 8-12mmHg
 - ▶ Mean Arterial Pressure > 65 mmHg
 - ▶ Urine Output >0.5 ml/kg/hr
 - ▶ Mixed Central Venous Saturation >70%
 - ▶ Or Mixed Venous Saturation >65% if Pulmonary Artery Catheter is used

Early Study: EGDT

Early Goal-Directed Therapy in Sepsis

Rivers *NEJM* 2001

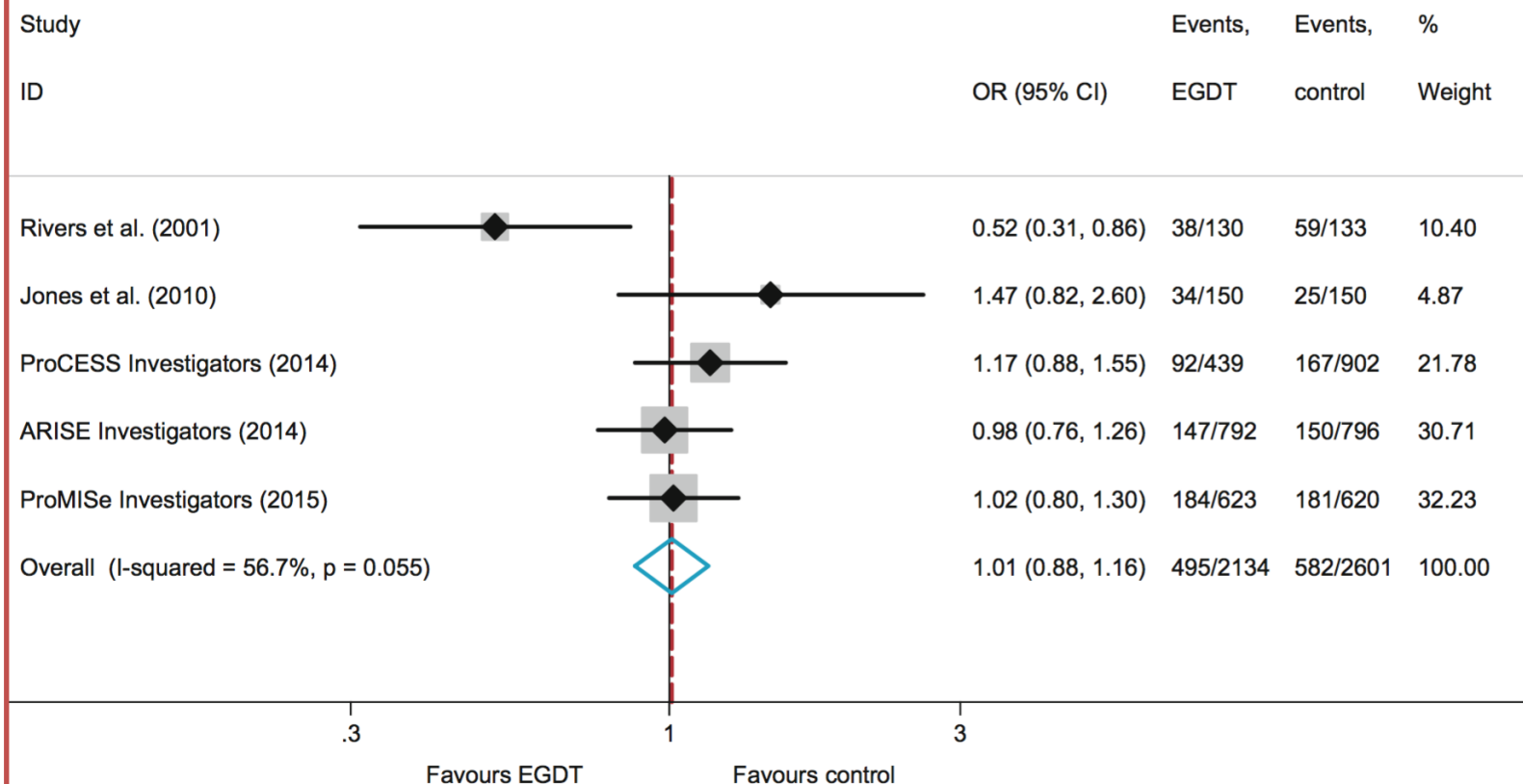
- ▶ Protocolized Resuscitation to Achieve targets in 6 hrs
- ▶ Single Centre Study
- ▶ Pending Result from Multi-Centre Studies on going in UK, USA and Australia



	EGDT	Control	P
n	130	133	
D28 Mortality	33.3%	49.2%	0.01
D60 Mortality	44.3%	56.9%	0.03

A systematic review and meta-analysis of early goal-directed therapy for septic shock: the ARISE, ProCESS and ProMiSe Investigators

A Primary mortality outcome of each study



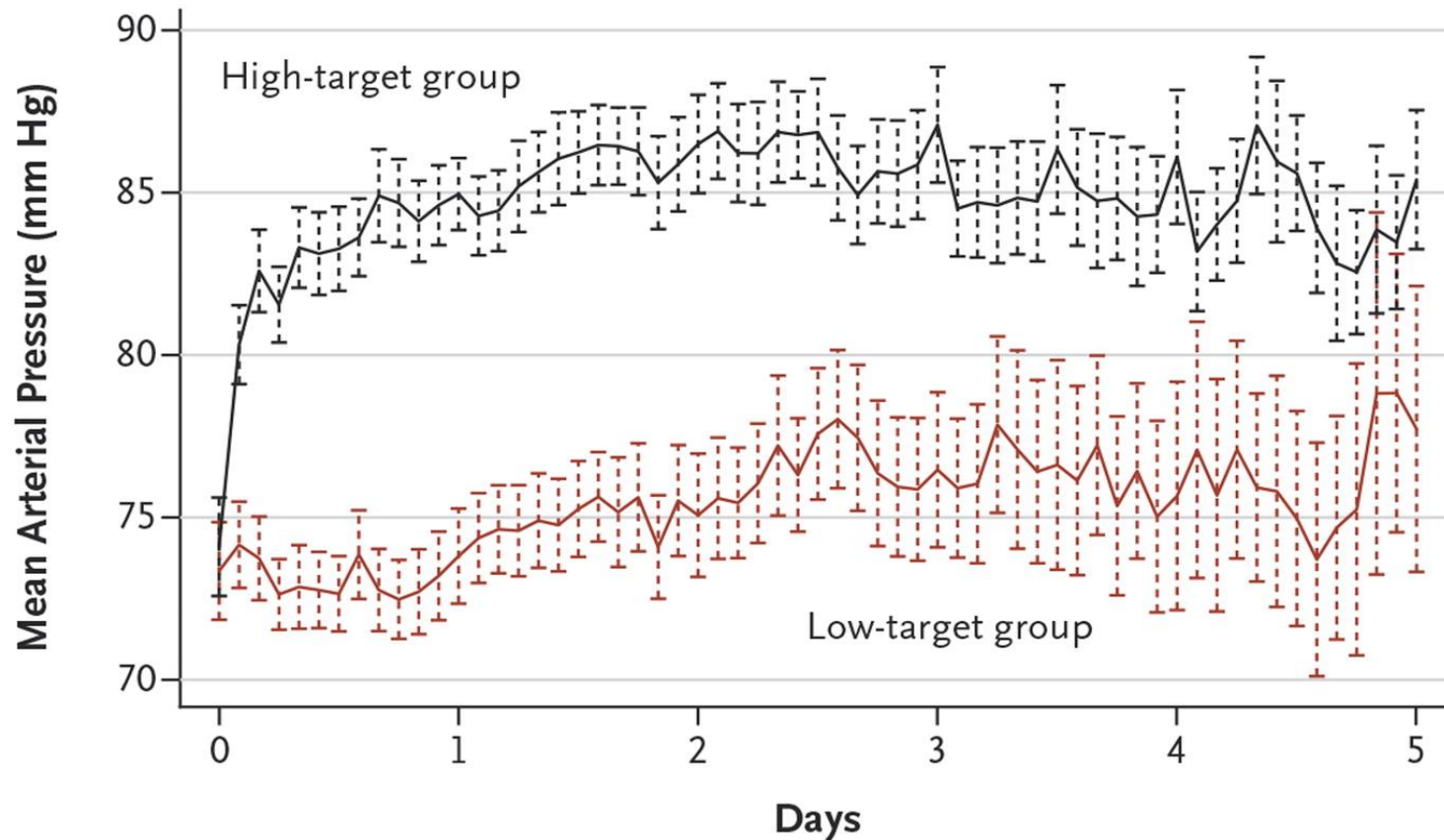
Does a Higher BP Help? N Engl J Med 2014;370:1583-93.

- ▶ **SEPSISPAM Study**: 29 French ICU
- ▶ Mar 2010-Dec2011
- ▶ Vasopressor dependence after 30ml/kg of IV Fluids

Target Blood Pressure	65-70mmHg	80-85mmHg	OR / p
N	388	388	
Mortality D28	34.0%	36.6%	1.07 (0.84-1.38) p=0.57
Rate of New AF	2.8%	6.7%	0.02
Renal Replacement Rx in Pt with Chronic HT	42.2%	31.7%	0.046

SEPSISPAM Study

Mean Arterial Pressure during the 5-Day Study Period.



Asfar P et al. N Engl J Med 2014;370:1583-1593.

A higher blood pressure is not associated with a better outcome

Which Fluid to Give?

Crystalloid	Colloids
• Normal Saline	Natural:
• Ringer's Lactate	• Albumin, Plasma Fractions
• Other Solutions	Synthetics:
	• Gelatin based (Gelofusine)
	• Starch based (Voluven)

SAFE Study (*NEJM* 2004)

- ▶ 4% albumin vs normal saline multi - centre RCT
- ▶ No difference in D28 mortality : 20.9% vs 21.1% $p=0.87$

Crystalloids vs Colloids in Sepsis

Study Group	Ref	Trial Protocol
CHEST Investigators and the Australian and New Zealand Intensive Care Society Clinical Trials Group	<i>N Engl J Med</i> 2012; 367:1901-1911	6% HES (130/0.4, Voluven) vs NS
Scandinavian Starch for Severe Sepsis/Septic Shock (6S) trial	<i>N Engl J Med</i> 2012; 367:124-134	6% HES 130/0.42 (Tetraspan) vs Ringers acetate



Crystalloids

Colloids

- For starch based colloids
- No survival benefits
- More Renal Failure



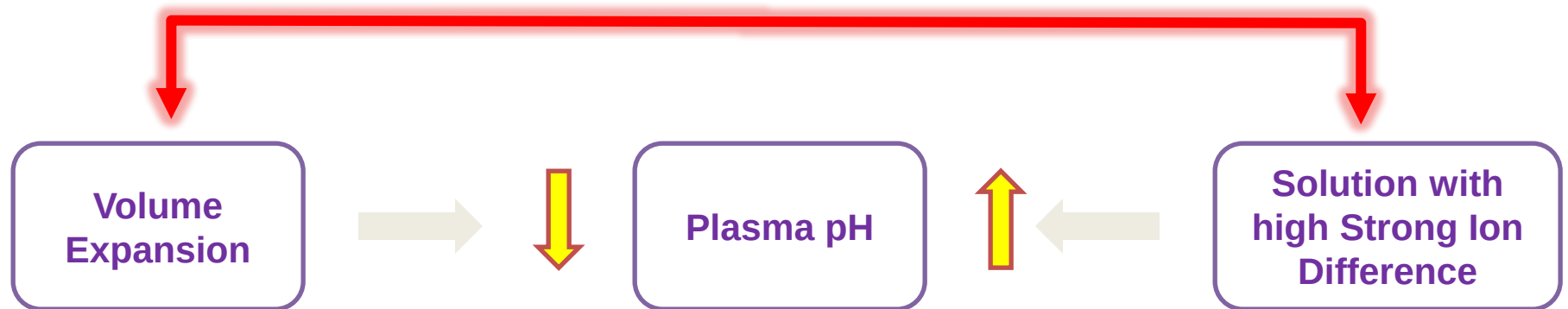
Results: 2 Major Randomized Trials

Study Group	Trial Protocol	N	D90 Mortality	Need Renal Replacement Rx
CHEST & ANZICS	6% HES 130/0.4, (Voluven)	3315	597 (18.0%)	235 (7.0%)
			RR =1.06 p =0.26	RR =1.21 P=0.04
	NS	3336	566 (17.0%)	196 (5.8%)
6S Trial	6% HES 130/0.42 (Tetraspan)	398	201 (51%)	87 (22%)
			RR =1.17 p =0.03	RR =1.35 p =0.04
	Ringers acetate	400	172 (43%)	65 (16%)



Balanced Solution

- Close to Physiological Electrolyte Composition
- Theoretical pH Neutral on massive infusion.



Comparison of Some Common IV Solutions

	Plasma	NS (0.9 NaCl)	Ringer's Lactate	Plasmalyte	Gelofusine
Na	140	154	130	140	154
K	0	0	4	5	0
Cl	100	154	109	98	125
Ca	2.2	0	1.5	0	0
Mg	1	0	1	1.5	0
HCO ₃	24	0	0	0	0
Lactate	1	0	28	0	0
Acetate	0	0	0	27	0
Gluconate	0	0	0	23	0

Is Hyperchloremia Bad?

ORIGINAL ARTICLE

Balanced Crystalloids versus Saline in Critically Ill Adults — A Systematic Review with Meta-Analysis

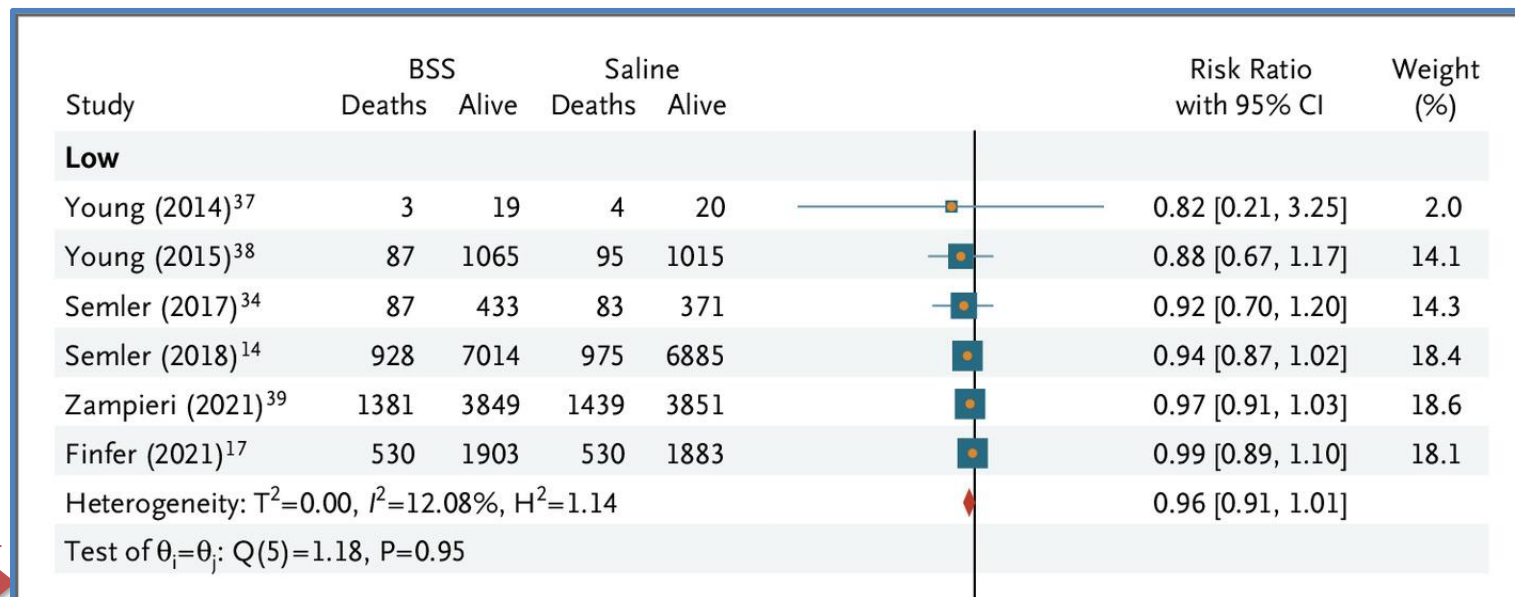
Naomi E. Hammond, Ph.D.^{1,2}, Fernando G. Zampieri, Ph.D.^{3,4}, Gian Luca Di Tanna, Ph.D.⁵, Tessa Garside, Ph.D.^{1,2}, Derick Adigbli, Ph.D.^{1,2}, Alexandre B. Cavalcanti, M.D. Ph.D.³, Flavia R. Machado, M.D., Ph.D.⁶, Sharon Micallef, B.N.¹, John Myburgh, Ph.D.^{1,7}, Mahesh Ramanan, M.Med.^{8,9}, Todd W. Rice, M.D.¹⁰, Matthew W. Semler, M.D.¹⁰, Paul J. Young, Ph.D.^{11,12}, Balasubramanian Venkatesh, M.D.^{1,13}, Simon Finfer, M.D.^{1,14}, and Anthony Delaney, Ph.D.^{1,2}

Drs. Hammond and Zampieri, as well as Drs. Finfer and Delaney, contributed equally to this article.

in Critically Ill Adults

Balanced vs NS	RR	95% CI
Mortality	0.96	0.91-1.01
Acute Kidney Injury	0.96	0.89-1.02
Needing Renal Replacement	0.95	0.81-1.11

also
low
bias



Blood Pressure vs Cardiac Output in Sepsis

- **NOT NECESSARY** targeting Supranormal Cardiac Output (Class I B)
- Choice of Drugs
 - **Inotropes:** Increases Cardiac Output
 - Mainly: Dobutamine (β_1 agonist)
 - Peripheral vasodilatation and may blood Pressure
 - **Vasopressors:**
 - Normally Vasoconstrictors to increase blood Pressure
 - Catecholamines:
 - Noradrenaline, Dopamine in High Dose, Phenylephrine
 - Vasopressin

Receptor	Action	Dopamine	Dobutamine	Epi	NorEpi
$\alpha 1$	Vasoconstriction	++ Hi Dose	0	+++	++++
$\beta 1$	Increase CO	+ Lo Dose	+++	+++	++

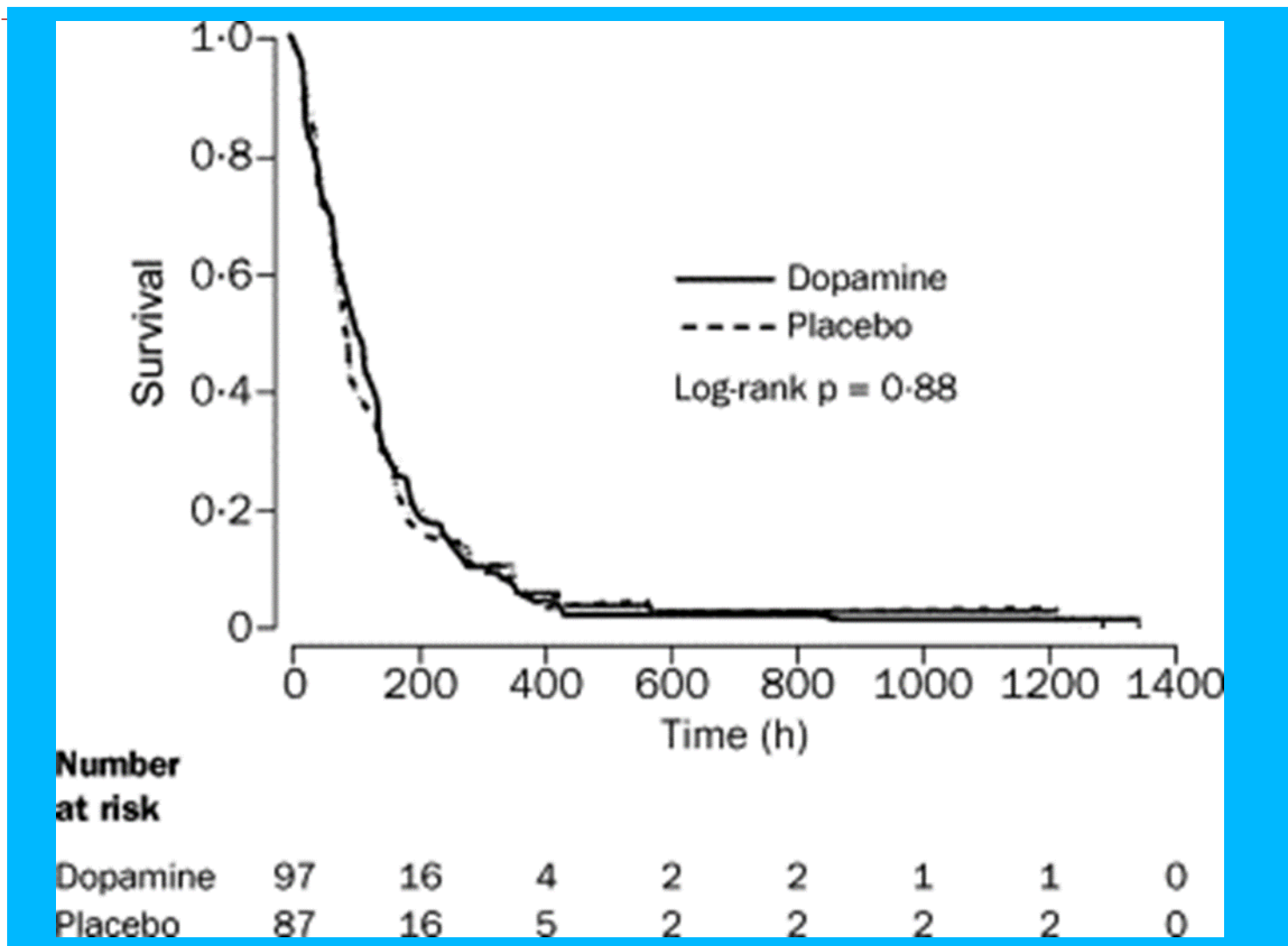
ANZIC Trial on Low Dose Dopamine

ANZIC Trial (Lancet 2000)

- ▶ ICU patients at risk of renal failure
- ▶ Dopamine (2 $\mu\text{g}/\text{kg}/\text{min}$) n=161
- ▶ Placebo n=163
- ▶ No difference in:
 - ▶ Peak serum creatinine
 - ▶ Patients requiring renal replacement therapy
 - ▶ ICU stay/hospital stay and mortality

“Renal Dose” Dopamine is “Out”

Dopamine ANZIC Study: Survival



Which Vasopressor?

Dopamine vs Norepinephrine

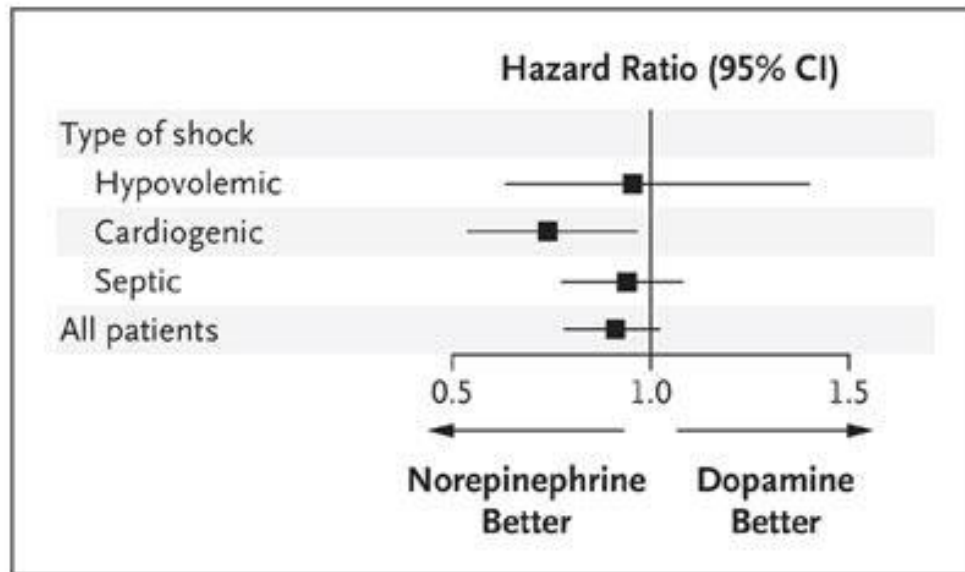
- ▶ SOAP II investigators
- ▶ Noradrenaline vs Dopamine multi-centre RCT *NEJM* 2010

	Dopamine	Noradrenaline	P
N	858	821	
D28 Mortality	52.5%	48.5%	0.10
Arrhythmias	24.1%	12.4%	<0.001
Discontinue for Arrhythmias	6.1%	1.6%	<0.001

- No difference in D28 Survival
- Dopamine Group had more patients who were discontinued from Rx for Arrhythmias

SOAP II Dopamine vs Norepinephrine

► Subgroup Analysis by Type of Shock



Noradrenaline
Recommended
Vasopressor for
septic shock

- Subgroup analysis:
 - In the subgroup with cardiogenic shock, Dopamine use was associated with Higher Mortality. ($p=0.03$)

Blood Glucose Control

- ▶ **Not** Necessary to Maintain Normal Blood Glucose
- ▶ Aim at Preventing Hyperglycemia of $>10\text{mmol/L}$
- ▶ Van de Berghe *NEJM* 2001
 - ▶ Surgical ICU, Mechanically Ventilated Patients

	Tight Glucose Control	Control Group	p
Target Glucose	4.4 – 6.0	10-11	
No. of Subjects	765	783	
ICU Mortality	4.6%	8.0%	0.04
Mortality of Group in ICU>5D	10.6%	20.2%	0.005

Sepsis: Other Studies on Glycemic Control (II)

► NICE Sugar Study *NEJM* 2009

► Multicentre Study

Tight Glucose Control Could Be Harmful!

	Tight Glucose	Control	p
Glucose Target	4.5 – 6.0	<10	
No. Subjects	3010	3012	
90D Mortality	27.5%	24.9%	0.02
Severe Hypoglycemia <2.2 mmol/L	6.8%	0.5%	<0.001

- No difference between Medical vs Surgical Patients
- No Difference in ICU LOS, Hospital LOS, Days on Ventilators and Renal Replacement

Lactate Clearance is Prognostic

- Lactate Clearance is associated with a better prognosis in patients with Sepsis
- 15 articles included
- RR of death with lactate clearance **0.38**
 - 95%CI 0.29-0.50

Lactate Clearance Is a Useful Biomarker for the Prediction of All-Cause Mortality in Critically Ill Patients: A Systematic Review and Meta-Analysis*

Zhongheng Zhang, MM; Xiao Xu, MB

Crit Care Med 2014

Lactate Clearance	Sensitivity	Specificity
All patients	0.75 (95% CI 0.58-0.87)	0.72 (95% CI 0.61-0.90)
ICU only	0.83 (95% CI 0.67-0.92)	0.67 (95% CI 0.59-0.75)

Evidence Based Medicine in ICU

**The Era of
Negative
Studies**

Study	Reference	
ANZICS Renal Dose Dopamine	Lancet 2000	
Albumin Study (SAFE)	NEJM 2004	-ve
VASST (NE vs Vasopressin)	NEJM 2008	-ve
CORTICUS	NEJM 2008	-ve
NICE-SUGAR	NEJM 2009	Intensive Group Worse
Dop vs NE SOAP II Investigators	NEJM 2010	-ve
PROWESS-SHOCK (Act Prot C vs P)	NEJM 2012	-ve
6 S Study HES 130/0.42 vs Ringer's	NEJM 2012	Starch worse



Mortality Trend of Patients with Septic Shock

- MICCOR Group Chest 2011
- US NIS database, Retrieved ICD 9 coding
- Year 2000-2007
- Incidence corrected to per 100,000 population

Parameter	2007 c.f 2000	P
No. severe Sepsis per 100,000	143 increase to 343	
No. organ failure	1.6 increase to 1.9	<0.001
Hospital LOS / Days	17.3 decrease to 14.9	
Mortality Rate	39.6 % decrease to 27.3%	
Discharge to Long Term Care Facility	27% increase to 35%	

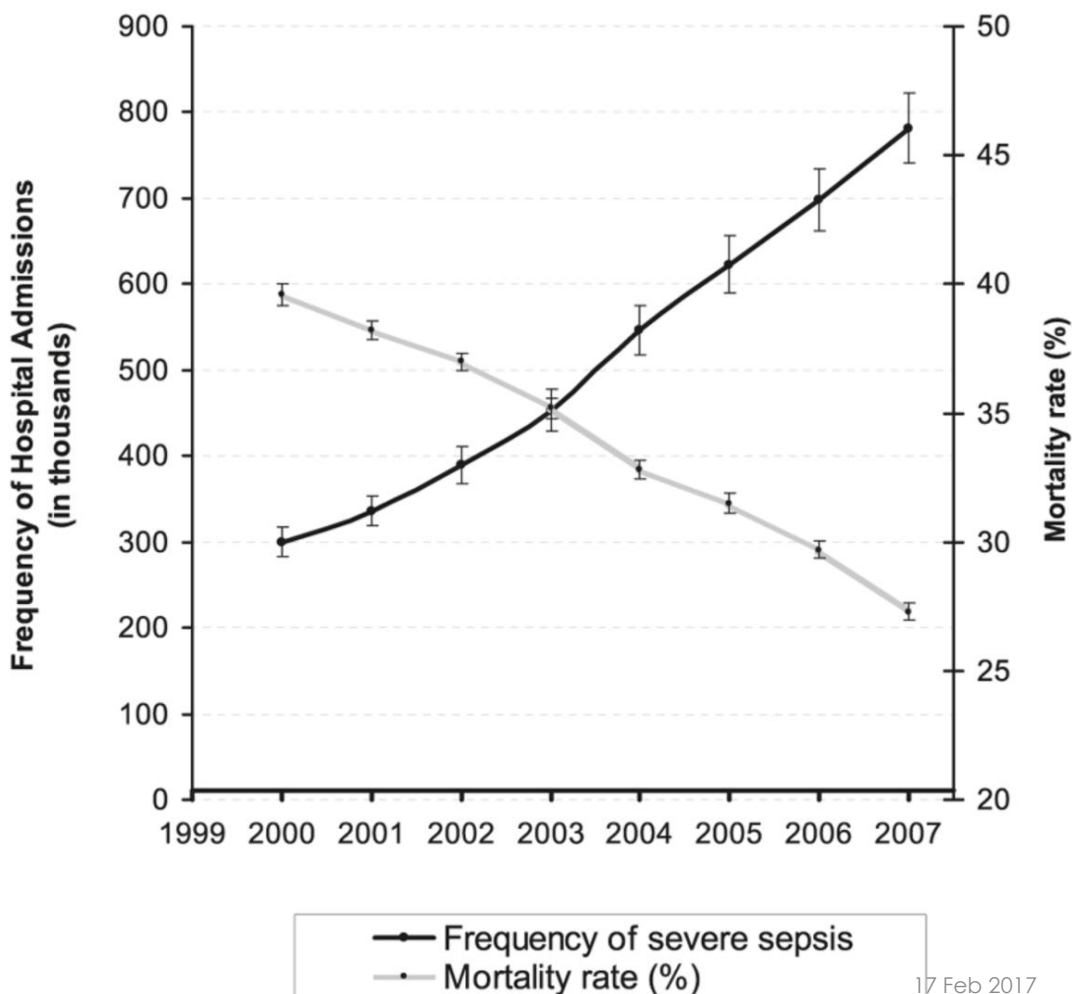


From: Nationwide Trends of Severe Sepsis in the 21st Century (2000-2007) National Trends of Severe Sepsis in 21st Century

Chest. 2011;140(5):1223-1231. doi:10.1378/chest.11-0352

Frequency of admission and mortality rates due to severe sepsis, 2000-2007. Bars represent SEM.

Figure Legend:



Summary

- ▶ Sepsis is a complicated immune process of the body in response to an infection
- ▶ Early Recognition of sepsis and effective anti-microbial therapy are important.
- ▶ Treatments that works theoretically need to be validated in randomized controlled trials.
- ▶ Specific drugs to modulate the process of sepsis in favour of survival is still lacking.

Thank You

JC Whole Class Session

**TELL US
WHAT
YOU THINK
THANK
YOU!**



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)